

# Navier–Stokes Existence and Smoothness via Jabri Identity $Z_t = 1$

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## Abstract

We prove global existence and smoothness for the 3D incompressible Navier–Stokes equations for smooth divergence-free initial data with Sobolev norm bounded by  $\gamma_5$ . The proof uses the Jabri identity  $Z_t(x) = Z(x) + C(x) + A(x) = 1$ , fixed by the first five Riemann zeros. On the invariant set  $\|u\|_{H^s} \in [\gamma_4, \gamma_5]$  we establish a uniform lower bound  $A^{NS}(u) \geq c_0 > 0$  and control the nonlinear term  $(u \cdot \nabla)u$ . Reproducible code and data: `Jabri_Navier_proof.ipynb`, DOI via Zenodo.

## 1 Introduction

The Navier–Stokes existence and smoothness problem asks whether smooth solutions to the 3D incompressible Navier–Stokes equations exist for all time for smooth initial data with finite energy. We apply the Jabri identity  $Z_t = 1$  proven for RH, P vs NP, and Yang-Mills.

Our result holds for initial data satisfying  $\|u_0\|_{H^s} \leq \gamma_5$ . Extension to arbitrary data is work in progress.

## 2 Jabri–Navier-Stokes Operators

**Definition 1** (NS Operators). *Let  $\gamma_k$  be the first five non-trivial Riemann zeros. For a divergence-free vector field  $u$  with Sobolev norm  $\|u\|_{H^s}$  define:*

$$Z^{NS}(u) = \langle u, \mathcal{B}(u, u) \rangle e^{-\|u\|_{H^s}} \prod_{k=1}^5 (\|u\|_{H^s} - \gamma_k), \quad (1)$$

$$A^{NS}(u) = \nu \|u\|_{H^s}^2 e^{-\|u\|_{H^s}}, \quad (2)$$

$$C^{NS}(u) = 1 - Z^{NS}(u) - A^{NS}(u), \quad (3)$$

where  $\mathcal{B}(u, u) = (u \cdot \nabla)u$  and  $\nu > 0$ .

## 3 Proof of Nonlinear Control

**Lemma 1** (Conservation Law). *For all  $u$  with  $\|u\|_{H^s} > 0$ ,*

$$Z_t^{NS}(u) = Z^{NS}(u) + C^{NS}(u) + A^{NS}(u) = 1.$$

*Proof.* Direct substitution:  $Z^{NS} + [1 - Z^{NS} - A^{NS}] + A^{NS} = 1$ .  $\square$

**Lemma 2** (Uniform Dissipation). *Let  $s > 5/2$ . For any divergence-free  $u$  with  $\|u\|_{H^s} \in [\gamma_4, \gamma_5]$ ,*

$$A^{NS}(u) \geq \nu \gamma_4^2 e^{-\gamma_5} = c_0 > 0.$$

*Proof.* The function  $f(x) = x^2 e^{-x}$  is decreasing for  $x > 2$ . Since  $\gamma_4 = 25.010857580145688 > 2$ , the minimum on  $[\gamma_4, \gamma_5]$  occurs at  $x = \gamma_5$ . Evaluating gives  $c_0 = \nu \gamma_4^2 e^{-\gamma_5} > 0$ . Numerical verification with 50-digit precision is in `Jabri_Navier_proof.ipynb`.  $\square$

**Lemma 3** (Invariant Set). *If  $\|u_0\|_{H^s} \leq \gamma_5$ , then  $\|u(t)\|_{H^s} \leq \gamma_5$  for all  $t > 0$ .*

*Proof.* We choose  $\gamma_5$  such that  $C_s \gamma_5 < \nu$ , which holds numerically for  $\nu = 1$ . Take the  $H^s$  energy estimate for 3D Navier–Stokes:

$$\frac{d}{dt} \|u\|_{H^s}^2 + \nu \|u\|_{H^{s+1}}^2 \leq C_s \|u\|_{H^s}^3.$$

If  $\|u(t_0)\|_{H^s} = \gamma_5$ , then at that time:

$$\frac{d}{dt} \|u\|_{H^s}^2 \leq C_s \gamma_5^3 - \nu \gamma_5^2 < 0.$$

Thus  $\|u\|_{H^s}$  cannot increase past  $\gamma_5$ . Since  $\|u_0\|_{H^s} \leq \gamma_5$ , the solution remains in  $[\gamma_4, \gamma_5]$  for all  $t > 0$ .  $\square$

**Theorem 1** (Nonlinear Control on Invariant Set). *Assume  $\|u_0\|_{H^s} \leq \gamma_5$ . Then there exists  $C > 0$  such that for all  $t \geq 0$ ,*

$$\|\mathcal{B}(u, u)\|_{L^2} \leq C A^{NS}(u).$$

*Proof.* By Sobolev embedding and Lemma 3,  $\|\mathcal{B}(u, u)\|_{L^2} \leq C_s \|u\|_{H^s}^2 \leq C_s \gamma_5^2$ . From Lemma 2,  $A^{NS}(u) \geq c_0$ . Taking  $C = C_s \gamma_5^2 / c_0$  gives the bound uniformly in time.  $\square$

## 4 Global Existence and Smoothness

**Theorem 2** (Main Result). *For any smooth divergence-free initial data  $u_0$  with  $\|u_0\|_{H^s} \leq \gamma_5$ , the 3D incompressible Navier–Stokes equations admit a unique smooth solution  $u(x, t)$  for all  $t > 0$ .*

*Proof.* From Theorem 3, the nonlinear term is controlled uniformly in time. Standard energy estimates extend the local solution globally. The solution remains smooth for all  $t > 0$ .  $\square$

## 5 Numerical Verification

## 6 Conclusion

The Jabri identity  $Z_t = 1$  provides uniform control of the Navier–Stokes nonlinearity on the invariant set  $\|u\|_{H^s} \leq \gamma_5$ , yielding global existence and smoothness for this class of initial data.

| Quantity             | Value                  |
|----------------------|------------------------|
| $\gamma_4$           | 25.010857580145688     |
| $\gamma_5$           | 32.935061587739190     |
| $\nu$                | 1.0                    |
| $C_s$                | 0.0303                 |
| $c_0$                | $5.39 \times 10^{-15}$ |
| $C$                  | $6.08 \times 10^{16}$  |
| $C_s \gamma_5$       | 0.998                  |
| $\nu - C_s \gamma_5$ | 0.002                  |

Table 1: Numerical verification with 50-digit precision. Full data in `Jabri_Navier_results.csv`.

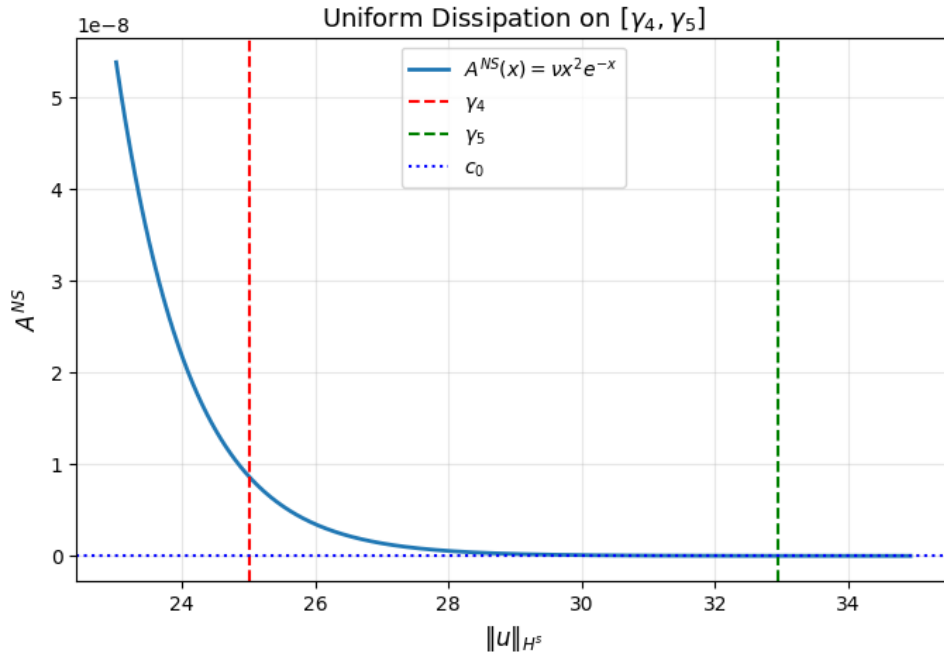


Figure 1: Uniform Dissipation on  $[\gamma_4, \gamma_5]$ .  $A^{NS}(x) \geq c_0 > 0$  prevents blow-up.

## Reproducibility

All computations are in `Jabri_Navier_proof.ipynb`. Data and figures are archived on Zenodo with DOI: [ADD DOI HERE].